

# HTML – Introduction

## What is HTML?

- **HTML** stands for **HyperText Markup Language**.
- It is the **standard language** used to create and design web pages.
- HTML is not a programming language, it is a **markup language**.
- A markup language uses **tags** to define the structure and elements of a webpage.

## Why HTML is important?

- It is the **backbone of every website**.
- Provides the **basic structure** of a webpage.
- All web technologies (like CSS, JavaScript, etc.) work **along with HTML**.
- Every browser (Chrome, Firefox, Edge, etc.) can read and display HTML documents.

## Features of HTML

1. **Simple & Easy to Learn** – Beginners can quickly understand and use it.
2. **Platform Independent** – Works on all operating systems (Windows, Mac, Linux).
3. **Supports Multimedia** – Can display text, images, audio, video, and animations.
4. **Hyperlinking** – Connects one page to another using **links**.
5. **Extensible** – Can be combined with CSS and JavaScript for advanced features.

## Structure of HTML Document

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Webpage</title>
</head>
<body>
  <h1>Hello, World!</h1>
  <p>This is my first webpage.</p>
</body>
</html>
```

### Explanation:

- `<!DOCTYPE html>` → Defines the document type (HTML5).
- `<html>` → Root element of the webpage.
- `<head>` → Contains meta information (title, styles, etc.).
- `<title>` → Title shown on the browser tab.
- `<body>` → Contains the actual content shown on the webpage.

## Advantages of HTML

- Free and easy to use.
- Supported by all web browsers.
- Helps in creating structured documents.
- Can integrate with CSS (for styling) and JavaScript (for interactivity).

## History of HTML:

- HTML was created by **Tim Berners-Lee** in **1991**.
- It was developed to share and structure information on the **World Wide Web (WWW)**.

- Over time, many versions of HTML have been released to improve features and support new technologies.

## Versions of HTML

1. **HTML 1.0 (1991):**
  - The first version, very basic, limited features.
2. **HTML 2.0 (1995):**
  - Standardized version, introduced forms and tables.
3. **HTML 3.2 (1997):**
  - Added support for scripting (JavaScript), applets, and more styling.
4. **HTML 4.01 (1999):**
  - Most popular older version, supported CSS, scripting, and multimedia.
5. **XHTML (2000):**
  - A stricter version of HTML, followed XML rules.
6. **HTML5 (2014 – Present):**
  - Current version.
  - Supports audio, video, canvas, SVG, and mobile-friendly websites.
  - Provides modern features for web apps.

## Basic HTML Tags

### 1. Document Tags

- `<!DOCTYPE html>` → Declares HTML5 document type.
- `<html>` → Root element of an HTML page.

### 2. Head Section Tags

- `<head>` → Contains metadata (information about the page).
- `<title>` → Defines the title of the webpage (shown in browser tab).
- `<meta>` → Provides metadata like character set, description, keywords.
- `<link>` → Used to connect external files (like CSS).

### 3. Body Section Tags

- `<body>` → Contains all visible content of the webpage.

### 4. Heading Tags

- `<h1>` to `<h6>` → Headings, where `<h1>` is the largest and `<h6>` is the smallest.

#### Example:

```
<h1>Main Heading</h1>
<h2>Sub Heading</h2>
```

## HTML Text Formatting Tags:

Text formatting tags are used to **change the appearance of text** (bold, italic, underline, etc.).

### 1. Bold Text

- `<b>` → Makes text **bold** (just for style).
- `<strong>` → Makes text **bold** and also indicates **importance** (semantic).

```
<p>This is <b>bold</b> text.</p>
```

<p>This is <strong>important</strong> text.</p>

## 2. Italic Text

- <i> → Makes text *italic* (just for style).
- <em> → Emphasizes text in *italic* (semantic, useful for screen readers).

<p>This is <i>italic</i> text.</p>

<p>This is <em>emphasized</em> text.</p>

## 3. Underline & Strike

- <u> → Underlines text.
- <s> or <strike> → Shows text as deleted (not recommended, old).
- <del> → Marks deleted text (semantic).
- <ins> → Marks inserted (underlined) text.

<p>This is <u>underlined</u> text.</p>

<p>This is <s>wrong</s> but <ins>correct</ins>.</p>

## 4. Superscript & Subscript

- <sup> → Superscript (above text line).
- <sub> → Subscript (below text line).

<p>2<sup>3</sup> = 8</p>

<p>H<sub>2</sub>O is water</p>

## 5. Small & Big Text

- <small> → Displays smaller text.
- <big> → Displays bigger text (deprecated in HTML5, use CSS).

<p>This is <small>small</small> text.</p>

<p>This is <big>big</big> text.</p>

## 6. Marked / Highlighted Text

- <mark> → Highlights text.

<p>This is <mark>highlighted</mark> text.</p>

## 7. Quote & Code Formatting

- <q> → Short quotation (with quotes).
- <blockquote> → Long quotation (indented).
- <code> → For code snippets.
- <pre> → Preformatted text (keeps spaces & line breaks).

<p>He said <q>Hello World!</q></p>

<blockquote>This is a long quoted text.</blockquote>

<pre>

Line 1

Line 2 (keeps spaces)

</pre>

<p>Use <code>printf()</code> in C.</p>

# HTML Lists

HTML provides different list types to organize items:

- **Unordered List (<ul>)** → Bulleted list
- **Ordered List (<ol>)** → Numbered/lettered list
- **(Definition List <dl>, <dt>, <dd>)**

## 1. Unordered List (<ul>)

- Displays items with **bullets** (• by default).
- Each list item is written inside <li> (list item).

### Example:

```
<ul>
  <li>Apple</li>
  <li>Mango</li>
  <li>Banana</li>
</ul>
```

Output:

- Apple
- Mango
- Banana

### Common Attributes of <ul>

| Attribute | Values                                 | Description                                                     |
|-----------|----------------------------------------|-----------------------------------------------------------------|
| type      | disc (•), circle (○), square (■), none | Defines bullet style (not supported in HTML5, use CSS instead). |

Example:

```
<ul type="square">
  <li>Item 1</li>
  <li>Item 2</li>
</ul>
```

## 2. Ordered List (<ol>)

- Displays items in a **numbered/lettered sequence**.
- Each item inside <li>.

### Example:

```
<ol>
  <li>Step 1</li>
  <li>Step 2</li>
  <li>Step 3</li>
</ol>
```

Output:

1. Step 1

2. Step 2
3. Step 3

## Common Attributes of <ol>

| Attribute | Values                                                                        | Description                    |
|-----------|-------------------------------------------------------------------------------|--------------------------------|
| type      | 1 (default numbers), A (A, B, C), a (a, b, c), I (I, II, III), i (i, ii, iii) | Defines numbering style        |
| start     | Any number                                                                    | Specifies starting value       |
| reversed  | Boolean (no value needed)                                                     | Displays list in reverse order |

## HTML Table Tags

Tables in HTML are created using the `<table>` element. They are used to display data in rows and columns.

### 1. Table Structure Tags

| Tag                          | Description                                    |
|------------------------------|------------------------------------------------|
| <code>&lt;table&gt;</code>   | Defines a table                                |
| <code>&lt;tr&gt;</code>      | Table row                                      |
| <code>&lt;th&gt;</code>      | Table header cell (bold & centered by default) |
| <code>&lt;td&gt;</code>      | Table data cell                                |
| <code>&lt;caption&gt;</code> | Adds a title/caption for the table             |
| <code>&lt;thead&gt;</code>   | Groups the header content                      |
| <code>&lt;tbody&gt;</code>   | Groups the body content                        |
| <code>&lt;tfoot&gt;</code>   | Groups the footer content                      |

### 2. Common Attributes of <table>

| Attribute   | Description                        | Example                                     |
|-------------|------------------------------------|---------------------------------------------|
| border      | Sets table border                  | <code>&lt;table border="1"&gt;</code>       |
| cellpadding | Space <b>inside</b> a cell         | <code>&lt;table cellpadding="10"&gt;</code> |
| cellspacing | Space <b>between</b> cells         | <code>&lt;table cellspacing="5"&gt;</code>  |
| width       | Width of table                     | <code>&lt;table width="80%"&gt;</code>      |
| align       | Aligns table (left, center, right) | <code>&lt;table align="center"&gt;</code>   |

| Attribute | Description                                               | Example                        |
|-----------|-----------------------------------------------------------|--------------------------------|
| bbgcolor  | Background color (old HTML, not recommended now, use CSS) | <table<br>bgcolor="lightblue"> |

### 3. Common Attributes of <tr>, <th>, <td>

| Attribute | Used in          | Description                              |
|-----------|------------------|------------------------------------------|
| align     | <tr>, <th>, <td> | Aligns content (left, center, right)     |
| valign    | <tr>, <th>, <td> | Vertical alignment (top, middle, bottom) |
| colspan   | <th>, <td>       | A cell spans <b>multiple columns</b>     |
| rowspan   | <th>, <td>       | A cell spans <b>multiple rows</b>        |
| bbgcolor  | <tr>, <th>, <td> | Background color of a cell/row           |

### 4. Example Table with Attributes

```
<!DOCTYPE html>
<html>
<head>
  <title>Table Example</title>
</head>
<body>

<table border="1" cellpadding="8" cellspacing="5" width="70%"
align="center">
  <caption><b>Student Marks</b></caption>

  <thead>
    <tr bgcolor="lightgray">
      <th>Roll No</th>
      <th>Name</th>
      <th colspan="2">Marks</th>
    </tr>
  </thead>

  <tbody>
    <tr>
      <td>101</td>
      <td>Amit</td>
      <td>85</td>
      <td>90</td>
    </tr>
    <tr>
      <td>102</td>
      <td>Neha</td>
      <td colspan="2" align="center">Absent</td>
    </tr>
    <tr>
      <td rowspan="2">103</td>
      <td>Ravi</td>
      <td>78</td>
      <td>88</td>
    </tr>
    <tr>
      <td>Priya</td>
      <td>92</td>
      <td>95</td>
    </tr>
  </tbody>
</table>
```

```

        </tr>
    </tbody>

    <tfoot>
        <tr bgcolor="lightyellow">
            <td colspan="4" align="center">Result: Passed</td>
        </tr>
    </tfoot>
</table>

</body>
</html>

```

## Semantic vs Non-Semantic Text Formatting Tags

### 1. What is Semantic HTML?

- **Semantic tags** have **meaning** – they describe the role or purpose of the text.
- Example: `<strong>` → means "important text", not just bold.
- Helps **search engines (SEO)** and **screen readers** understand the content better.

### 2. What is Non-Semantic HTML?

- **Non-semantic tags** only define **style/appearance** of the text.
- They do not provide meaning, only change how the text looks.
- Example: `<b>` → makes text bold but does not say it is important.

### 3. Key Differences

| Feature                   | Semantic Tags                                                                                               | Non-Semantic Tags                                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Meaning</b>            | Carry meaning (importance/emphasis)                                                                         | No meaning, only style                                                                            |
| <b>Accessibility</b>      | Helps screen readers & SEO                                                                                  | Not useful for accessibility                                                                      |
| <b>Examples</b>           | <code>&lt;strong&gt;</code> , <code>&lt;em&gt;</code> , <code>&lt;del&gt;</code> , <code>&lt;ins&gt;</code> | <code>&lt;b&gt;</code> , <code>&lt;i&gt;</code> , <code>&lt;u&gt;</code> , <code>&lt;s&gt;</code> |
| <b>Usage</b>              | For meaningful/important text                                                                               | For styling only                                                                                  |
| <b>Preferred in HTML5</b> | Yes, recommended                                                                                            | Not recommended (use CSS instead)                                                                 |

### 4. Examples

#### Non-Semantic:

```

<p>This is <b>bold</b> text.</p>
<p>This is <i>italic</i> text.</p>

```

Text is bold/italic, but no special meaning.

#### Semantic:

```

<p>This is <strong>important</strong> text.</p>
<p>This is <em>emphasized</em> text.</p>

```

Text is bold/italic **and** browsers, search engines, and screen readers know that it's important/emphasized.

# HTML Forms

## What is a Form?

- An **HTML form** is used to collect user input.
- Example: Login page, signup form, feedback form.
- All form elements are written inside the `<form>` tag.

## 1. The `<form>` Tag

### Important Attributes of `<form>`

| Attribute | Values              | Description                                                                                                                                                                                           |
|-----------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| action    | URL (server script) | Where to send form data (e.g., PHP file, API).                                                                                                                                                        |
| method    | get, post           | HTTP method: <ul style="list-style-type: none"><li>• <b>GET</b> – Data visible in URL (less secure, used for search).</li><li>• <b>POST</b> – Data hidden, secure (used for login, signup).</li></ul> |

## 2. Common Form Elements

### (a) Input Field `<input>`

Used for text, passwords, checkboxes, radio buttons, etc.

#### Attributes of `<input>`:

- `type` → Defines input type (text, password, email, number, date, file, checkbox, radio, submit, reset, etc.).
- `name` → Name of input (sent to server).
- `value` → Default value.
- `placeholder` → Hint text inside field.
- `required` → Field must be filled before submission.
- `readonly` → Makes field non-editable.
- `disabled` → Field disabled (not sent to server).
- `maxlength` / `minlength` → Restrict input length.
- `min` / `max` → Set numeric/date range.

Example:

```
<input type="text" name="username" placeholder="Enter your name" required>
<input type="password" name="pass" maxlength="12">
<input type="checkbox" name="agree"> I Agree
<input type="radio" name="gender" value="male"> Male
<input type="radio" name="gender" value="female"> Female
```

### (b) Textarea `<textarea>`

Multi-line input field.

```
<textarea name="message" rows="5" cols="30" placeholder="Enter your message"></textarea>
```

### (c) Select (Dropdown) `<select>` with `<option>`

```
<select name="course">
  <option value="html">HTML</option>
  <option value="css">CSS</option>
  <option value="js">JavaScript</option>
</select>
```

Attributes: `multiple` (for multiple selection), `selected` (default option).

### (d) Button

- `<button>` or `<input type="button">` → Normal button.
- `<input type="submit">` → Submit form.
- `<input type="reset">` → Reset form values.

Example:

```
<input type="submit" value="Register">
<input type="reset" value="Clear">
<button type="button">Click Me</button>
```

### (e) File Upload

```
<input type="file" name="resume">
```

### (f) Label `<label>`

Links text to a form element for accessibility.

```
<label for="uname">Username:</label>
<input type="text" id="uname" name="username">
```

### Tag `<optgroup>`

- Used to **group related options** inside a dropdown.
- Helps organize long dropdowns.

### Syntax:

```
<select name="fruits">
  <optgroup label="Citrus">
    <option value="orange">Orange</option>
    <option value="lemon">Lemon</option>
  </optgroup>
  <optgroup label="Berries">
    <option value="strawberry">Strawberry</option>
    <option value="blueberry">Blueberry</option>
  </optgroup>
</select>
```

Output:

- Citrus
  - Orange
  - Lemon
- Berries
  - Strawberry
  - Blueberry

### 3. Complete Example Form

```
<!DOCTYPE html>
<html>
<head>
  <title>Registration Form</title>
</head>
<body>

  <form action="submit.php" method="post" autocomplete="on">

    <label for="uname">Username:</label>
    <input type="text" id="uname" name="username" placeholder="Enter name"
required><br><br>

    <label for="pass">Password:</label>
    <input type="password" id="pass" name="password" minlength="6"
required><br><br>

    Gender:
    <input type="radio" name="gender" value="male"> Male
    <input type="radio" name="gender" value="female"> Female<br><br>

    <label for="course">Select Course:</label>
    <select id="course" name="course">
      <option value="html">HTML</option>
      <option value="css">CSS</option>
      <option value="js">JavaScript</option>
    </select><br><br>

    <label for="msg">Message:</label><br>
    <textarea id="msg" name="message" rows="4" cols="30"></textarea><br><br>

    <label for="resume">Upload Resume:</label>
    <input type="file" id="resume" name="resume"><br><br>

    <input type="submit" value="Register">
    <input type="reset" value="Clear">

  </form>

</body>
</html>
```

## Linking & Navigation Tags in HTML

These tags are used to create links, connect resources, and navigate between different sections or pages of a website.

### 1. <a> (Anchor Tag)

The <a> tag is used to create a **hyperlink**. It can link to another webpage, section on the same page, email, or file.

#### Syntax:

```
<a href="https://example.com" target="_blank">Visit Example</a>
```

#### Used Attributes:

- **href** → Specifies the link destination (URL or file).
- **target** → Defines where to open the link.

- o `_self` (default, same tab)
  - o `_blank` (new tab)
- **download** → Used for downloading a file instead of opening it.

## 2. <nav> (Navigation Tag)

The `<nav>` tag defines a **navigation section** that contains links for moving within a website.

### Syntax:

```
<nav>
  <a href="index.html">Home</a>
  <a href="about.html">About</a>
  <a href="contact.html">Contact</a>
</nav>
```

## 3. <link> (Link Tag)

The `<link>` tag is used to connect external resources like **CSS files, icons, or fonts**. It is placed inside the `<head>` section.

### Syntax:

```
<link rel="stylesheet" href="style.css">
<link rel="icon" href="favicon.ico" type="image/x-icon">
```

### Used Attributes:

- **rel** → Defines relationship (e.g., stylesheet, icon).
- **href** → Specifies the file path (CSS, favicon, etc.).
- **type** → Specifies file type (e.g., text/css, image/x-icon).

## 4. <iframe> (Inline Frame)

The `<iframe>` tag is used to **embed another webpage** within the current page.

### Syntax:

```
<iframe src="https://example.com" width="600" height="400"></iframe>
```

### Used Attributes:

- **src** → URL of the embedded page.
- **width & height** → Set size of the frame.

# Media Tags in HTML

Media tags are used to display and control **images, audio, and video** on a web page.

## 1. <img> (Image Tag)

Used to insert images in a webpage.

### Syntax:

```

```

### Most Used Attributes:

- **src** → Path/URL of the image.
- **alt** → Alternative text (shown if image is not loaded).
- **width** → Sets image width.
- **height** → Sets image height.

## 2. <audio> (Audio Tag)

Used to play sound/audio files.

### Syntax:

```
<audio controls autoplay loop>
  <source src="song.mp3" type="audio/mpeg">
  Your browser does not support the audio element.
</audio>
```

### Most Used Attributes:

- **src** → Direct audio file path (if no <source> is used).
- **controls** → Adds play/pause, volume controls.
- **autoplay** → Plays audio automatically when page loads.
- **loop** → Repeats audio automatically.
- **muted** → Starts audio muted.

## 3. <video> (Video Tag)

Used to embed video files in a webpage.

### Syntax:

```
<video width="600" height="400" controls autoplay loop muted>
  <source src="movie.mp4" type="video/mp4">
  Your browser does not support the video tag.
</video>
```

### Most Used Attributes:

- **src** → Direct video path (if no <source> is used).
- **width & height** → Set video dimensions.
- **controls** → Adds play, pause, volume, fullscreen options.
- **autoplay** → Starts video automatically.
- **loop** → Repeats the video automatically.
- **muted** → Starts video muted.
- **poster** → Sets an image to display before video starts.

## 4. <source> (Source Tag)

Defines multiple file formats for <audio> and <video>.  
This ensures cross-browser compatibility.

### Syntax:

```
<video controls>
  <source src="movie.mp4" type="video/mp4">
  <source src="movie.ogg" type="video/ogg">
```

</video>

### Most Used Attributes:

- **src** → Path of media file.
- **type** → MIME type of the file (video/mp4, audio/mpeg, etc.).

## 5. <track> (Track Tag)

Used inside <video> for **subtitles, captions, or descriptions**.

### Syntax:

```
<video controls>
  <source src="movie.mp4" type="video/mp4">
  <track src="subtitles.vtt" kind="subtitles" srclang="en" label="English">
</video>
```

### Most Used Attributes:

- **src** → File containing subtitles (usually .vtt).
- **kind** → Type of text track:
  - subtitles, captions, descriptions, chapters, metadata.
- **srclang** → Language of the track (e.g., en, fr).
- **label** → Title shown in the video player for subtitles.

## 6. <embed> (Embed Tag)

The <embed> tag is used to **embed external content** (like audio, video, PDF, Flash, or another application) directly into a webpage.

### Syntax:

```
<embed src="movie.mp4" width="600" height="400">
```

### Most Used Attributes:

- **src** → Path/URL of the file to embed.
- **type** → MIME type of the file (e.g., video/mp4, application/pdf).
- **width** → Sets width of the embedded content.
- **height** → Sets height of the embedded content.

Note: <embed> is not as flexible as <video> or <audio>. It is often used for **PDF files** and some special cases.

## 7. <object> (Object Tag)

The <object> tag is a **container for external resources** like PDFs, multimedia, or applications. It is more powerful than <embed>.

### Syntax:

```
<object data="file.pdf" type="application/pdf" width="600" height="400">
  Your browser does not support embedded PDF.
</object>
```

### Most Used Attributes:

- **data** → Path of the file to be embedded.
- **type** → MIME type (like `application/pdf`, `video/mp4`).
- **width & height** → Size of embedded content.
- **name** → Gives a name to the object (used in scripting).

## 8. <param> (Parameter Tag)

The `<param>` tag is used **inside** `<object>` to pass parameters to plugins or embedded applications.

### Syntax:

```
<object data="movie.mp4" type="video/mp4" width="600" height="400">
  <param name="autoplay" value="true">
  <param name="loop" value="true">
</object>
```

### Most Used Attributes:

- **name** → Name of the parameter.
- **value** → Value of the parameter.

## Semantic Tags in HTML

Semantic tags clearly describe the meaning of the content inside them. Unlike `<div>` or `<span>` (non-semantic), these tags tell both the browser and developers about the **role of the content**.

### 1. <header>

Represents the **top section** of a page or a section (logo, navigation, heading, etc.).

### Syntax:

```
<header>
  <h1>My Website</h1>
  <nav>
    <a href="#">Home</a>
    <a href="#">About</a>
  </nav>
</header>
```

### 2. <footer>

Represents the **bottom section** of a page or section (copyright, contact info, links).

### Syntax:

```
<footer>
  <p>&copy; 2025 My Website</p>
</footer>
```

### 3. <main>

Defines the **main content area** of the page. Only one `<main>` per page is allowed.

### Syntax:

```
<main>
  <article>
    <h2>Article Title</h2>
    <p>This is main content.</p>
  </article>
</main>
```

#### 4. <article>

Represents **independent content** like a blog post, news article, or forum post.

##### Syntax:

```
<article>
  <h2>Blog Post</h2>
  <p>This is a standalone article.</p>
</article>
```

#### 5. <section>

Defines a **thematic grouping of content** (chapters, sections of a page).

##### Syntax:

```
<section>
  <h2>Services</h2>
  <p>We provide web development and design.</p>
</section>
```

#### 6. <aside>

Represents **side content** (sidebar, ads, related info).

##### Syntax:

```
<aside>
  <h3>Related Links</h3>
  <ul>
    <li><a href="#">HTML Basics</a></li>
  </ul>
</aside>
```

#### 7. <figure> and <figcaption>

Used to group media content (images, diagrams) with a caption.

##### Syntax:

```
<figure>
  
  <figcaption>Beautiful Nature</figcaption>
</figure>
```

#### 7. <details>

Creates a **collapsible section** that can be expanded or collapsed by the user.

##### Syntax:

```
<details open>
```

```
<summary>Read More</summary>
<p>This is hidden content that shows when expanded.</p>
</details>
```

## 9. <time>

Represents a **time, date, or both** in machine-readable format. Useful for calendars, events, blog posts.

### Syntax:

```
<p>Event starts at <time datetime="2025-08-22T10:00">10:00 AM</time>.</p>
```

### Most Used Attributes:

- **datetime** → Machine-readable date/time.

## 10. <address>

Represents **contact information** for a person or organization.

### Syntax:

```
<address>
  Contact us: <a href="mailto:info@example.com">info@example.com</a>
</address>
```

## 11. <abbr>

Defines an **abbreviation or acronym** with a full form as a tooltip.

### Syntax:

```
<p>The <abbr title="World Health Organization">WHO</abbr> was founded in
1948.</p>
```

# Block-level vs Inline Elements in HTML

## Block-level Elements

- **Definition:** Block-level elements always start on a new line and take up the full width of their parent container.
- **Behavior:** Browsers automatically add a line break before and after block elements.
- **Use:** They are used to structure the layout into larger sections.
- **Examples:**

- ✓ <div>
- ✓ <p>
- ✓ <h1> to <h6>
- ✓ <section>, <article>
- ✓ <ul>, <ol>, <li>
- ✓ <form>, <table>

### Example:

```
<p>This is a paragraph.</p>
<p>This is another paragraph.</p>
```

Both <p> elements appear on separate lines.

## Inline Elements

- **Definition:** Inline elements do not start on a new line. They only take up as much width as their content.
- **Behavior:** They sit inside a line along with text or other inline elements.
- **Use:** They are used to style or format parts of text.
- **Examples:**

- ✓ <span>
- ✓ <a>
- ✓ <strong>, <em>
- ✓ <img>
- ✓ <label>
- ✓ <b>, <i>

### Example:

```
<p>This is <span>inline text</span> inside a paragraph.</p>
```

The word *inline text* appears within the same line of the paragraph.

## Metadata & Scripting Tags in HTML

These tags provide **information about the webpage**, connect external resources, and allow embedding of **CSS or JavaScript**. Most of them are used inside the <head> section of an HTML document.

### 1. <head>

The <head> tag contains **metadata** (information about the document) that is not displayed directly on the page.

#### Syntax:

```
<head>  
  <title>My Website</title>  
  <meta charset="UTF-8">  
</head>
```

### 2. <title>

Specifies the **title of the webpage**, shown in the browser tab and used by search engines.

#### Syntax:

```
<title>HTML Notes</title>
```

### 3. <meta>

Provides **metadata** like description, keywords, author, charset, and viewport settings.

#### Syntax:

```
<meta charset="UTF-8">  
<meta name="description" content="Learn HTML with notes">
```

```
<meta name="keywords" content="HTML, CSS, JavaScript">
<meta name="author" content="Neeraj Katheriya">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<meta http-equiv="refresh" content="10; url=https://example.com">
<meta http-equiv="X-UA-Compatible" content="IE=edge">
```

### Most Used Attributes:

- **charset** → Defines character encoding (e.g., UTF-8).
- **name** → Type of metadata (description, keywords, author, viewport).
- **content** → Value of the metadata.
- **http-equiv** → For HTTP headers (e.g., refresh, X-UA-Compatible).

## 5. <style>

Used to write **internal CSS** inside an HTML document.

### Syntax:

```
<style>
  body { background-color: lightblue; }
</style>
```

## 6. <script>

Used to write **JavaScript code** or link an external JS file.

### Syntax (Internal JS):

```
<script>
  alert("Hello World!");
</script>
```

### Syntax (External JS):

```
<script src="app.js"></script>
```

### Most Used Attributes:

- **src** → Path of external JS file.
- **type** → Type of script (text/javascript, optional in HTML5).

## What is SEO?

**SEO** stands for **Search Engine Optimization**.

It is the process of **improving a website** so that it appears **higher in search engine results** (like Google, Bing, Yahoo).

When someone searches for a keyword related to your site, good SEO helps your website appear on the **first page** instead of being hidden on later pages.

### Why SEO is Important?

- Increases **website visibility**
- Brings **organic (free) traffic**
- Builds **trust & credibility**
- Better **user experience**

- Helps compete with other websites

## Extra HTML Tags

### 1. <dialog>

- Defines a popup dialog box.
- **Attributes:**
  - `open` → makes dialog visible.

#### Example:

```
<dialog open>
  <p>This is a dialog box</p>
</dialog>
```

### 2. <menu>

- Defines a list of commands or options (rarely used).

### 3. <progress> Tag

- Represents the **progress of a task**.
- Mostly used for **loading bars or task completion**.

#### Attributes:

- `value` → current progress (number).
- `max` → maximum value (default is 1).

#### Example:

```
<label for="task">Task Progress:</label>
<progress id="task" value="60" max="100"></progress>
```

Shows progress as **60% completed**.

### 4. <meter> Tag

- Represents a **measurement within a known range** (e.g., disk usage, temperature, score).

#### Attributes:

- `value` → current value.
- `min` → minimum value.
- `max` → maximum value.
- `low, high` → define ranges.
- `optimum` → ideal value.

#### Example:

```
<p>Disk Usage: <meter value="4" min="0" max="10">4 out of 10</meter></p>
```

## 5. <dl>, <dt>, <dd> (Definition List)

- Used for **glossaries, FAQs, or descriptions**.

### Tags:

- <dl> → Definition List (container).
- <dt> → Definition Term.
- <dd> → Definition Description.

### Example:

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>

  <dt>CSS</dt>
  <dd>Cascading Style Sheets</dd>
</dl>
```

## 6. <fieldset> Tag

- The <fieldset> tag is used to **group related input elements** in a form.
- The browser automatically draws a **border box** around the group.
- It makes forms more **organized and easier to read**.

## 7. <legend> Tag

- The <legend> tag is used to provide a **title or caption** for the <fieldset>.
- It is displayed at the **top-left corner** of the fieldset box.
- Only **one <legend> is allowed** inside a <fieldset>.

### Example:

```
<form>
  <fieldset>
    <legend>Personal Information</legend>
    Name: <input type="text"><br>
    Age: <input type="number"><br>
  </fieldset>

  <fieldset>
    <legend>Contact Details</legend>
    Email: <input type="email"><br>
    Phone: <input type="tel"><br>
  </fieldset>
</form>
```

## 8. <svg> Tag (Scalable Vector Graphics)

- Used to draw **shapes, graphics, and icons** directly in HTML.
- Scales without losing quality.

### Example:

```
<svg width="400" height="250" viewBox="0 0 400 250" aria-label="SVG
Shapes">
  <!-- rectangle -->
  <rect x="10" y="10" width="120" height="80" fill="#ffa94d"
stroke="#222" stroke-width="2"></rect>
```

```
<!-- circle -->
<circle cx="220" cy="50" r="40" fill="#74c0fc" stroke="#222" stroke-
width="2"></circle>

<!-- line -->
<line x1="10" y1="130" x2="390" y2="130" stroke="#fa5252" stroke-
width="4"></line>

<!-- triangle -->
<polygon points="300,30 360,110 240,110" fill="#69db7c" stroke="#222"
stroke-width="2"></polygon>

<!-- text -->
<text x="10" y="200" font-size="24" fill="#2f9e44">Hello SVG</text>
</svg>
```